AstraFlame: Physics-Informed Machine Learning for Spacecraft Fire Safety from NASA Microgravity Combustion Data

NASA Space Apps Challenge 2026 · Challenge: *Flame in Freefall: AI-Powered Fire Safety Insights from Microgravity Combustion Data*

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Abstract

NASA has proposed a thinner, oxygen-richer cabin atmosphere for exploration habitats (8.2 psia, 34% O_2), and its engineering assessments identify material flammability as the most critical hurdle in that environment. Of the 22 materials whose oxygen limits NASA measured (Hirsch et al. 2011), 20 would keep burning at 34% oxygen, compared with 12 in Earth air. Fire also behaves differently in low gravity, and NASA's microgravity evidence is spread across separate investigations, file formats and reports.

AstraFlame brings that evidence together. We built a reproducible dataset of 850 verified records from nine NASA microgravity investigations (FLEX, BASS, BASS-II, SPICE, SAME, SAME-R and Saffire I–III), plus 52 Earth-gravity reference rows. Every processed file can be rebuilt byte-for-byte from the NASA files, and 890 values are re-checked against NASA on every rebuild. On this data we trained five models that answer five fire-safety questions. For the three outputs that feed the safe-cabin map (flame spread, flame extinction and droplet burning rate), we use a physics-informed Gaussian process: a physics law, a Gaussian-process correction kept only when validation supports it, and an ISS-test offset. The two smoke outputs use tuned ensembles. Every score comes from nested grouped cross-validation, in which whole ISS tests are held out, and is compared with a noise ceiling estimated from NASA's own replicate runs.

Task (NASA source)	Final model	Held-out score	Share of the noise ceiling
Droplet burning rate (FLEX)	Physics-informed GP	R^2 0.71	98%
Smoke point (SPICE)	Tuned ensemble	R^2 0.89	89%
Flame spread (BASS-II)	Physics-informed GP	R^2 0.57	84%
Flame extinction (FLEX)	Physics-informed GP classifier	ROC-AUC 0.87, balanced accuracy 0.78	Brier 0.082 vs 0.114 baseline
Smoke particle size (SAME)	Tuned ensemble	R^2 0.43	51% (weakest task)

Every prediction comes with a 90% range, a warning when the input lies outside what NASA tested, and exact Shapley values that show which inputs drove the answer. Because the models know where they are unsure, AstraFlame also recommends where NASA's next ISS fire experiment would teach the most: a 3 mm methanol droplet at 32% O_2 and 144 kPa, and flame spread at 21% O_2 with 7.5 cm/s airflow. The dashboard combines these models with a low-cost pressure chamber and a live pressure sensor, material gauges, a pressure-to-gravity dial and NASA citations for every number.

We state our limits openly. The material oxygen limits come from Earth-gravity tests, so they are only a first estimate for 8.2 psia. Extinction errors rise above 21% oxygen, where NASA ran few tests. With 52–256 rows per task, the confidence intervals are wide. AstraFlame is decision support for habitat designers choosing materials and cabin air before launch, not a certification tool.

Keywords: microgravity combustion, spacecraft fire safety, exploration atmosphere, physics-informed Gaussian process, uncertainty quantification, grouped cross-validation, NASA Physical Sciences Informatics.
Code and data: github.com/prachyabiswas/NASA-Space-Apps-2026 · NASA data: psi.nasa.gov (CC0-1.0), ntrs.nasa.gov.